

MapShop Tanzania Backend API

Overview

RESTful API backend for MapShop Tanzania - a map-based e-commerce platform for smart local shopping and delivery.

Features

- User authentication (JWT)
- Shop and product management
- Order processing with real-time tracking
- GPS-based location services
- Delivery rider management
- Push notifications
- SMS and email notifications
- Payment integration

Technology Stack

- Django 5.0
- Django REST Framework
- PostgreSQL with PostGIS
- Redis (caching & WebSocket)
- Celery (async tasks)
- Google Maps API
- Twilio (SMS)
- SendGrid (Email)
- JWT Authentication

Installation

Prerequisites

- Python 3.10+

- PostgreSQL 14+ with PostGIS
- Redis 6+
- Google Maps API Key

Setup

1. Clone the repository

```
git clone https://github.com/mln4h/mapshop-tanzania.git
cd mapshop-tanzania/backend

# create the virtual environment
python3 -m venv venv
source venv/bin/activate # On Windows: venv\Scripts\activate

# install the dependencies
pip install -r requirements.txt

## SETUP DATABASE
# Create PostgreSQL database with PostGIS
sudo -u postgres createdb mapshop_db
sudo -u postgres psql -c "CREATE EXTENSION postgis;" mapshop_db

# Run migrations
python3 manage.py makemigrations
python3 manage.py migrate

##CREATE THE SUPERUSER

python manage.py createsuperuser

## RUNSERVER
python3 manage.py runserver
python3 manage.py runserver 0.0.0.0:8000
cd ..
```

```
### INSTALLATION SETUP
postgresql 16..
postGIS 3..
Redis server 7...
GEOS and GDAL (geos-3.12.1
GDAL 3.8.4, released 2024/02/08
)
-django redis
```